

Ordering Integers on a Number Line

PhET Number Line: Integers · PhET

Years 7-8	~30 minutes	Mathematics · Number, Ratios & Financial Maths
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/number-line-integers>

Curriculum links: AC9M8N04 · number

Learning intentions

- I can place positive and negative integers correctly on a number line.
- I can compare and order integers using real-world contexts like temperature and bank balance.
- I can use the four operations with integers to solve simple problems.

Getting started

Open PhET Number Line: Integers and choose a context screen such as 'Bank' or 'Temperature'. In each round, drag the marker (or type a value) to where you think the number belongs on the number line, then reveal the correct position to check yourself.

Tasks

1. Choose the 'Bank' context. If your balance is -\$15, are you in debt or in credit? Explain how you know from the sign.

2. Place five given values on the number line before revealing the answer each time. Record how many you got exactly right.

Value	Your placement	Correct? (Y/N)

3. Switch to the 'Temperature' context. Which is colder: -8°C or -3°C? Explain how the number line helps you answer this, even though -8 looks like a 'bigger' number without the sign.

4. Order these values from smallest to largest without using the simulation first, then check your order using the number line: -12, 5, -3, 0, 8.

5. Explain why -20 is less than -5, even though 20 is greater than 5. Use the number line to support your explanation.

6. A submarine is at an altitude of -45 m and a plane is at an altitude of 1200 m. What is the difference in altitude between them? Check your reasoning using the number line.

7. A student says 'negative numbers are just smaller versions of positive numbers.' Use at least one example from this simulation to explain why this idea is incomplete.

Extension (optional)

Your bank balance is -\$40. You make a deposit of \$25 and then a withdrawal of \$10. What is your final balance? Show each step on a number line and explain whether you are in debt at the end.
